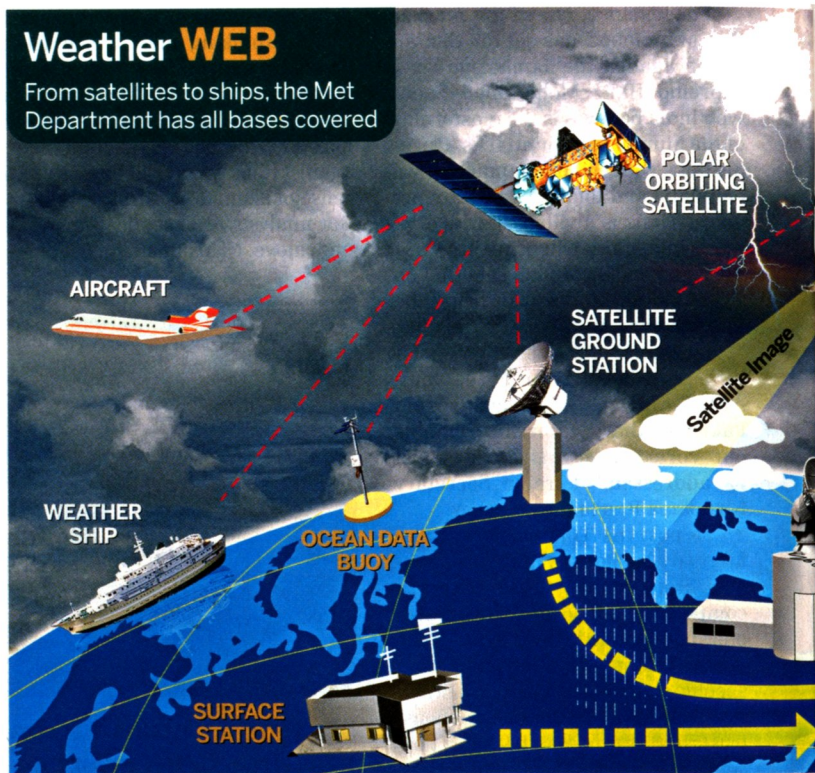


By Amarnath K. Menon

He is customarily candid but is circumspect about his current task. Air Vice-Marshall (Retd) Ajit Tyagi is more certain about blue skies than dark clouds. Occupying the hot seat in Delhi's Mausam Bhavan, the India Meteorological Department (IMD) director-general knows well that he has to offer a convincing answer to a difficult question: how much will it rain this year? Tyagi, on his part, is able to make only an inspired guess based on interpreting a pool of data. His initial forecast is that it will be a 'normal' monsoon. It is a poor consolation, considering the projection of a 'near-normal' monsoon in 2009 ended in abject failure.

In his defence, Tyagi turns to the IMD records. Only thrice in the last 110 years was rainfall deficient, at least 10 per cent below normal, observed in successive years: in 1904-05, 1965-66 and 1986-87. Weather observations made against a historical set of parameters and modelled on a super-computer to predict the monsoon sig-



Chasing THE RAINBOW

The Met Department's monsoon forecasting effort struggles with complex atmospheric dynamics to come up with accurate predictions

nal good rainfall this year. This explains why Tyagi almost dismisses the idea of bad monsoon two years in a row. He points out that out of about 20 droughts since 1901, 17 were followed by near-normal rainfall.

There is a need for accurate forecasts not only in advance but throughout the season on the spread and distribution of the rains. It helps farmers plan, plant and nurture crops with greater confidence. Every six out of 10 Indians, or over 700 million people,

depend on agriculture and related services for their livelihood. While agriculture accounts for only 20 per cent of the GDP, the fact is rural purchasing power is critical for overall growth. India's resilience in the face of the 2009 meltdown is only because of rural resurgence. Therefore, dependable weather forecast is of paramount importance.

Monsoon forecasting is a complex dynamic which is not fully understood. It is a challenge which the IMD

took up in June 1886 and is still far from getting it fully right. India, in fact, is the first country to embark on operational seasonal forecasting.

Tyagi has high hopes this time, thanks to two crucial indicators—the absence of warm (El Nino) or cold (La Nina) environments in the tropical Pacific. Although the El Nino and La Nina cycle does not occur with strict periodicity, they have a recurring pattern with an irregular cycle of two to seven years. The current El Nino had